

Partial Global-Convergence Results and a Contraction Counterexample for Anderson Mixing

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Abstract

We study ordinary, unsafeguarded type-II Anderson mixing in exact arithmetic. Three results are proved. First, for an affine map, exact Anderson mixing with arbitrary finite memory is globally linearly convergent whenever the linear part of the corresponding damped fixed-point iteration is uniformly contractive; no bound on the Anderson coefficients is needed. Second, in one dimension, ordinary Anderson mixing of depth one is globally linearly convergent for every q -contraction with $q < 1/3$, under the standard Picard initialization. Third, contraction alone is not sufficient in the full range $q < 1$: we construct an explicit piecewise-affine contraction with Lipschitz constant $(1 + \sqrt{5})/4$ for which ordinary Anderson(1) has an exact nontrivial cycle of period four. These results constitute partial progress only. They do not resolve the general global-convergence question for ordinary nonlinear Anderson mixing in Euclidean space.

Status of the open problem. This note gives *partial progress, not a solution*. It proves global convergence for two restricted classes and gives a counterexample showing that global contractivity of the fixed-point map, by itself, cannot yield a general theorem for ordinary Anderson mixing.

1 Introduction and scope

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and consider the fixed-point equation

$$g(x) = x.$$

Anderson mixing, also called Anderson acceleration, uses a least-squares combination of recent residual differences to accelerate the basic fixed-point iteration; see [1, 2]. Li and Bao observe that a general global convergence theory for ordinary Anderson mixing remains unavailable even in Euclidean space [5]. Their globally convergent RRAM method is a regularized modification of ordinary Anderson mixing, whereas their unregularized RAM method is analyzed locally. Classical nonlinear analyses also commonly impose local hypotheses or an a priori bound on the Anderson coefficients; see, for example, [3].

The associated online problem statement asks for assumptions and parameter choices under which Anderson mixing converges from arbitrary initial points. Read literally, such a formulation is too weak: choosing memory $m = 0$ reduces the method to damped Picard iteration, whose global convergence for contractions is immediate. We therefore interpret the substantive question as follows:

Find nontrivial classes of maps for which *ordinary, unsafeguarded least-squares Anderson mixing with memory $m \geq 1$* is globally convergent, or identify obstructions to such a theorem.

The present note makes three contributions under this interpretation.

- (i) For affine maps, a simple residual identity gives global linear convergence for arbitrary finite memory under a uniform norm condition on the damped linear part.
- (ii) For scalar q -contractions with $q < 1/3$, depth-one Anderson mixing is the secant method applied to a strongly decreasing residual, and a global contraction estimate follows.
- (iii) An explicit scalar contraction with Lipschitz constant

$$q_{\text{cyc}} = \frac{1 + \sqrt{5}}{4} \approx 0.80901699$$

produces an exact period-four orbit under ordinary Anderson(1), despite standard Picard initialization.

Thus the general question is not solved. The positive results are restricted, and the counterexample shows that the apparently natural hypothesis “ g is a global contraction” is insufficient without a stronger quantitative restriction or additional structure. We make no claim of priority for the individual observations; the purpose is to give a self-contained, mathematically verified account of what is actually proved.

2 Ordinary type-II Anderson mixing

Define the residual

$$r(x) := g(x) - x.$$

Fix a maximum memory $m \geq 1$, and let $m_k := \min\{m, k\}$. For $m_k \geq 1$, define

$$\begin{aligned} X_k &:= [x_{k-m_k+1} - x_{k-m_k}, \dots, x_k - x_{k-1}], \\ R_k &:= [r(x_{k-m_k+1}) - r(x_{k-m_k}), \dots, r(x_k) - r(x_{k-1})]. \end{aligned}$$

At iteration k , ordinary type-II Anderson mixing chooses any exact least-squares minimizer

$$\gamma_k \in \arg \min_{\gamma \in \mathbb{R}^{m_k}} \|r(x_k) - R_k \gamma\|_2, \tag{1}$$

sets

$$\bar{x}_k := x_k - X_k \gamma_k, \quad \bar{r}_k := r(x_k) - R_k \gamma_k, \tag{2}$$

and updates

$$x_{k+1} := \bar{x}_k + \beta_k \bar{r}_k. \tag{3}$$

When $m_k = 0$, the matrices are empty and the convention is

$$x_{k+1} = x_k + \beta_k r(x_k).$$

Throughout, “ordinary” means that the least-squares problem is not regularized or constrained and that no restart, acceptance test, or fallback step is added.

3 Affine maps

We first record an elementary global theorem for the exact unmodified scheme. Its proof uses no coefficient bound.

Theorem 3.1 (Affine residual contraction). *Let*

$$g(x) = Ax + b, \quad B := I - A,$$

and run (1)–(3) with any finite maximum memory m , exact least squares, and arbitrary real damping parameters β_k . Suppose

$$\rho := \sup_{k \geq 0} \|I - \beta_k B\|_2 < 1. \quad (4)$$

Then B is nonsingular, g has the unique fixed point

$$x^* = B^{-1}b,$$

and every initial point satisfies

$$\|r(x_k)\|_2 \leq \rho^k \|r(x_0)\|_2, \quad (5)$$

$$\|x_k - x^*\|_2 \leq \|B^{-1}\|_2 \rho^k \|r(x_0)\|_2. \quad (6)$$

The conclusion does not require bounded Anderson coefficients.

Proof. If $Bv = 0$ for some nonzero v , then

$$(I - \beta_k B)v = v$$

for every k , contradicting (4). Hence B is nonsingular and the fixed point is unique.

Since

$$r(x) = b - Bx,$$

every residual difference equals minus B times the corresponding iterate difference. Therefore

$$R_k = -BX_k. \quad (7)$$

Using (7) and (2),

$$\begin{aligned} r(\bar{x}_k) &= b - B(x_k - X_k \gamma_k) \\ &= r(x_k) + BX_k \gamma_k \\ &= r(x_k) - R_k \gamma_k = \bar{r}_k. \end{aligned} \quad (8)$$

Consequently,

$$\begin{aligned} r(x_{k+1}) &= b - B(\bar{x}_k + \beta_k \bar{r}_k) \\ &= r(\bar{x}_k) - \beta_k B \bar{r}_k \\ &= (I - \beta_k B) \bar{r}_k. \end{aligned}$$

The vector $\gamma = 0$ is feasible in (1), so exact least squares gives

$$\|\bar{r}_k\|_2 \leq \|r(x_k)\|_2.$$

It follows that

$$\|r(x_{k+1})\|_2 \leq \rho \|r(x_k)\|_2.$$

Iteration proves (5). Finally,

$$r(x) = -B(x - x^*),$$

so

$$\|x_k - x^*\|_2 \leq \|B^{-1}\|_2 \|r(x_k)\|_2,$$

which yields (6). $\square$

Corollary 3.2 (Affine contractions). *Suppose $\|A\|_2 \leq q < 1$ and*

$$0 < \underline{\beta} \leq \beta_k \leq 1.$$

Then Theorem 3.1 applies with

$$\rho \leq 1 - \underline{\beta}(1 - q) < 1.$$

In particular, undamped ordinary Anderson mixing, $\beta_k = 1$, is globally convergent for every affine Euclidean contraction and every finite memory m .

Proof. For $0 \leq \beta \leq 1$,

$$\|(1 - \beta)I + \beta A\|_2 \leq 1 - \beta + \beta q = 1 - \beta(1 - q).$$

Since $(1 - \beta)I + \beta A = I - \beta B$, the result follows from Theorem 3.1. $\square$

Remark 3.3 (Scope of the affine theorem). Condition (4) already makes the underlying damped affine fixed-point map contractive. The nontrivial point is that exact least-squares Anderson mixing cannot destroy this residual contraction, regardless of the size of its coefficients. Linear Anderson acceleration has a substantial existing theory; see, for example, [4]. The theorem above is included because the proof is short and directly verifies the claimed global bound, not as a claim that the linear case was previously unresolved.

4 Scalar contractions with $q < 1/3$

We now turn to a genuinely nonlinear result for the unmodified method. In one dimension, exact Anderson mixing of depth one coincides with the secant method for the residual.

Lemma 4.1 (Residual slope bounds). *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a q -contraction with $0 \leq q < 1$, and let $r = g - I$. For all distinct $u, v \in \mathbb{R}$,*

$$-(1 + q) \leq \frac{r(u) - r(v)}{u - v} \leq -(1 - q). \quad (9)$$

In particular, r is strictly decreasing and its unique zero is the unique fixed point x^ of g .*

Proof. Write

$$\frac{r(u) - r(v)}{u - v} = \frac{g(u) - g(v)}{u - v} - 1.$$

Contractivity implies

$$-q \leq \frac{g(u) - g(v)}{u - v} \leq q,$$

which gives (9). Strict decrease follows because the upper endpoint is negative. Existence and uniqueness of x^* also follow from the Banach fixed-point theorem. $\square$

Theorem 4.2 (Global convergence of scalar Anderson(1)). *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a q -contraction with*

$$0 \leq q < \frac{1}{3}.$$

Start from an arbitrary x_0 , take the standard Picard initialization

$$x_1 := g(x_0),$$

and, from iteration $k = 1$ onward, run exact ordinary Anderson mixing with depth $m = 1$. If a fixed point is encountered, terminate. Otherwise every least-squares denominator is nonzero, the damping parameters β_k for $k \geq 1$ are immaterial, and

$$|x_{k+1} - x^*| \leq \chi |x_k - x^*|, \quad \chi := \frac{2q}{1-q} < 1, \quad k \geq 1. \quad (10)$$

Consequently,

$$|x_k - x^*| \leq q \chi^{k-1} |x_0 - x^*|, \quad k \geq 1, \quad (11)$$

with the case $q = 0$ understood as immediate termination after the Picard step.

Proof. The case $q = 0$ is immediate because a 0-contraction is constant. Assume $0 < q < 1/3$ and that x_0 is not fixed. Then $x_1 \neq x_0$.

Suppose inductively that $x_k \neq x_{k-1}$. Lemma 4.1 implies

$$r(x_k) - r(x_{k-1}) \neq 0.$$

The scalar least-squares problem therefore has the unique solution

$$\gamma_k = \frac{r(x_k)}{r(x_k) - r(x_{k-1})},$$

and its minimized residual is exactly zero:

$$\bar{r}_k = r(x_k) - (r(x_k) - r(x_{k-1}))\gamma_k = 0.$$

Thus the Anderson update is independent of β_k and equals the secant step

$$x_{k+1} = x_k - r(x_k) \frac{x_k - x_{k-1}}{r(x_k) - r(x_{k-1})}. \quad (12)$$

If x_k is not fixed, then the correction in (12) is nonzero, so $x_{k+1} \neq x_k$. This proves inductively that the iteration is well-defined until it reaches the fixed point.

Assume now that $x_k \neq x^*$. Define

$$s_k := \frac{r(x_k)}{x_k - x^*}, \quad d_k := \frac{r(x_k) - r(x_{k-1})}{x_k - x_{k-1}}.$$

By Lemma 4.1,

$$s_k, d_k \in [-(1+q), -(1-q)].$$

Equation (12) gives

$$x_{k+1} - x^* = \left(1 - \frac{s_k}{d_k}\right) (x_k - x^*).$$

Since s_k and d_k are both negative,

$$\frac{s_k}{d_k} \in \left[\frac{1-q}{1+q}, \frac{1+q}{1-q} \right].$$

Therefore

$$\begin{aligned} \left| 1 - \frac{s_k}{d_k} \right| &\leq \frac{1+q}{1-q} - 1 \\ &= \frac{2q}{1-q} = \chi, \end{aligned}$$

which proves (10). Finally,

$$|x_1 - x^*| = |g(x_0) - g(x^*)| \leq q|x_0 - x^*|,$$

and iteration of (10) yields (11). $\square$

Remark 4.3. The estimate proves global safety, not acceleration: the guaranteed factor $2q/(1-q)$ is larger than the Picard factor q . The threshold $q < 1/3$ is sufficient, not asserted to be necessary.

5 A scalar contraction with an exact period-four Anderson cycle

We next show that global contractivity alone does not guarantee convergence of ordinary Anderson(1). The construction is exact and one-dimensional.

Let

$$\phi := \frac{1 + \sqrt{5}}{2}, \quad q_{\text{cyc}} := \frac{\phi}{2} = \frac{1 + \sqrt{5}}{4} < 1.$$

Recall the identities

$$\phi^2 = \phi + 1, \quad \phi^{-1} = \phi - 1, \quad \phi^{-2} = 2 - \phi.$$

Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) := \begin{cases} -\frac{\phi}{2}, & x \leq -\phi^2, \\ \frac{x+1}{2}, & -\phi^2 \leq x \leq -\phi^{-2}, \\ -\frac{\phi}{2}x, & -\phi^{-2} \leq x \leq \frac{1}{2\phi}, \\ \frac{\phi}{2}x - \frac{1}{2}, & \frac{1}{2\phi} \leq x \leq 1, \\ \frac{1}{2\phi}, & x \geq 1. \end{cases} \quad (13)$$

Proposition 5.1 (The map is a contraction). *The function in (13) is continuous and globally q_{cyc} -Lipschitz. It is therefore a contraction and has the unique fixed point $x^* = 0$.*

Proof. The formulas agree at every breakpoint. For example,

$$\frac{-\phi^2 + 1}{2} = -\frac{\phi}{2}, \quad \frac{1 - \phi^{-2}}{2} = \frac{1}{2\phi},$$

and the remaining junctions are equally direct. The slopes of the five affine pieces are

$$0, \quad \frac{1}{2}, \quad -\frac{\phi}{2}, \quad \frac{\phi}{2}, \quad 0.$$

Hence the absolute slope is everywhere at most $\phi/2 = q_{\text{cyc}}$. A continuous piecewise-affine function with this slope bound is globally q_{cyc} -Lipschitz. Since $g(0) = 0$ and $q_{\text{cyc}} < 1$, the Banach fixed-point theorem gives uniqueness of the fixed point. $\square$

Theorem 5.2 (Exact period-four cycle). *Apply ordinary Anderson mixing of depth one to the map (13), using exact least squares and the standard initialization*

$$x_0 := 1, \quad x_1 := g(x_0) = \frac{1}{2\phi}.$$

Then the iterates form the nontrivial period-four cycle

$$1, \quad \frac{1}{2\phi}, \quad -\phi^2, \quad -\phi^{-2}, \quad 1, \quad \frac{1}{2\phi}, \quad \dots \quad (14)$$

In particular, ordinary Anderson(1) does not converge, for any choice of damping parameters after the initialization.

Proof. Write $r = g - I$ and set

$$x_0 = 1, \quad x_1 = \frac{1}{2\phi}, \quad x_2 = -\phi^2, \quad x_3 = -\phi^{-2}.$$

The values of g and r at these four points are

x	$g(x)$	$r(x) = g(x) - x$
$x_0 = 1$	$\frac{1}{2\phi}$	$\frac{\phi - 3}{2}$
$x_1 = \frac{1}{2\phi}$	$-\frac{1}{4}$	$-\frac{\sqrt{5}}{4}$
$x_2 = -\phi^2$	$-\frac{\phi}{2}$	$1 + \frac{\phi}{2}$
$x_3 = -\phi^{-2}$	$\frac{1}{2\phi}$	$\frac{3 - \phi}{2}$

For depth one, the exact Anderson update is the secant formula (12). The four successive residual secant slopes are

$$d_1 := \frac{r(x_1) - r(x_0)}{x_1 - x_0} = -\frac{\phi^{-2}}{2}, \quad (15)$$

$$d_2 := \frac{r(x_2) - r(x_1)}{x_2 - x_1} = -\frac{\phi}{2}, \quad (16)$$

$$d_3 := \frac{r(x_3) - r(x_2)}{x_3 - x_2} = -\frac{1}{2}, \quad (17)$$

$$d_4 := \frac{r(x_0) - r(x_3)}{x_0 - x_3} = -1. \quad (18)$$

All denominators are nonzero. Direct substitution, using the identities for ϕ , gives

$$\begin{aligned}x_1 - \frac{r(x_1)}{d_1} &= -\phi^2 = x_2, \\x_2 - \frac{r(x_2)}{d_2} &= -\phi^{-2} = x_3, \\x_3 - \frac{r(x_3)}{d_3} &= 1 = x_0, \\x_0 - \frac{r(x_0)}{d_4} &= \frac{1}{2\phi} = x_1.\end{aligned}$$

Thus the ordered pair of consecutive iterates returns to (x_0, x_1) after four Anderson steps, proving (14). At every scalar depth-one step the minimized residual $\bar{r}_k$ is zero, so the damping term in (3) vanishes; hence later damping choices cannot remove the cycle. $\square$

Corollary 5.3 (Contractivity alone is insufficient). *For every $q \in [q_{\text{cyc}}, 1)$, there exists a scalar map with Lipschitz constant at most q , a unique fixed point, and an initial point for which ordinary Anderson(1) with standard Picard initialization does not converge.*

Proof. The map (13) has Lipschitz constant q_{cyc} and is therefore also a q -contraction for every $q \geq q_{\text{cyc}}$. The nonconvergent orbit is given by Theorem 5.2. $\square$

Remark 5.4 (The remaining quantitative gap). Theorem 4.2 gives uniform global convergence for all scalar q -contractions when $q < 1/3$, whereas Theorem 5.2 gives failure at $q = (1 + \sqrt{5})/4$. This note does not determine what happens uniformly over the intermediate range

$$\frac{1}{3} \leq q < \frac{1 + \sqrt{5}}{4}.$$

6 What is and is not resolved

The results can be classified as follows.

1. The affine theorem is a complete global result for a restricted, structurally simple class. It applies to the ordinary unregularized method and permits arbitrary finite memory.
2. The scalar theorem is genuine nonlinear global convergence for ordinary Anderson(1), but only in one dimension and only under the strong quantitative condition $q < 1/3$.
3. The period-four example is a genuine obstruction: even in one dimension, a global contraction with a unique fixed point need not be globally convergent under ordinary Anderson(1).

None of these statements supplies a global theory for general nonlinear maps in $\mathbb{R}^n$. In particular, they do not prove global convergence of ordinary Anderson mixing under the broad assumptions contemplated by Li and Bao. Safeguarded, regularized, or fallback variants may be globally convergent, but they are different algorithms and do not by themselves settle the behavior of the ordinary unsafeguarded least-squares iteration. The mathematically accurate classification of this note is therefore:

partial progress, not a solution of the general open problem.

References

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